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### Method for controlling an aircraft capable of hovering and relative aircraft

#### Abstract

A method for controlling an aircraft capable of hovering is described, comprising a first engine; a second engine; at least one rotor; and a transmission interposed between the first and second engine and the rotor; the transmission comprises a first and a second inlet connected respectively to a first outlet member of the first engine and to a second outlet member of the second engine; the method comprises step i) of placing the in a first configuration, in which the first and second engine make available a first and a second power value; or in a second configuration, in which the first engine makes available a third power value greater than the first power value to the first inlet, and the second engine delivers a nil power value to the second inlet; the method also comprises, characterised in that it comprises the steps of ii) detecting a series of parameters associated with the operating conditions of the aircraft; and iii) enabling the transition of the aircraft from the first configuration to the second configuration, when the parameters assume respective first values.

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## **Background/Summary**

### **CROSS-REFERENCE TO RELATED APPLICATIONS**

(1) This Patent Application is a U.S. National Phase Application under 35 U.S.C. § 371 of International Patent Application No. PCT/IB2021/056978, filed on Jul. 30, 2021, which claims priority from European Patent Application No. 20189398.9 filed on Aug. 4, 2020, all of which are incorporated by reference, as if expressly set forth in their respective entireties.

### **TECHNICAL FIELD**

(2) The present invention relates to a method for controlling an aircraft capable of hovering and to the aircraft itself.

(3) In more detail, the aircraft is a helicopter or a convertiplane.

### **BACKGROUND ART**

(4) Helicopters in a known manner essentially comprise a fuselage, a main rotor rotatable around a first axis and arranged on a top portion of the fuselage, and an anti-torque rotor arranged at a tail end of the helicopter and rotatable around a second axis, transverse to the first axis.

(5) The helicopters of the known type also comprise an engine system and a transmission unit for transmitting the motion from an outlet shaft of the engine system to the main rotor.

(6) In multi-engine solutions, the engine system comprises at least one pair of engines, more precisely turbo-shaft engines, which are provided with respective outlet shafts connected to the transmission.

(7) At least in twin-engine helicopters, each “turbo-shaft” engine is oversized to ensure the necessary power in case of failure of the other turbo-shaft engine.

(8) Consequently, each turbo-shaft engine is capable of delivering a maximum power greater than that needed by the helicopter when both turbo-shaft engines are in operation.

(9) Said oversizing is suboptimal in terms of weight and fuel consumption of the turbo-shaft engines.

(10) In fact, at cruise speed in forward flight, the turbo-shaft engines deliver respective power levels approximately comprised between a value of 55% and a value of 80% of such maximum power value.

(11) At these power levels, the engine efficiency is much lower than the efficiency at power levels close to 100%, with higher specific fuel consumption.

(12) In other words, in general the efficiency of a turbo-shaft engine that operates at a level equal to 60% of the maximum deliverable power has a higher specific fuel consumption than when it delivers the maximum power.

- (13) This implies that, generally, on a helicopter, turbo-shaft engines have a suboptimal specific fuel consumption at cruise speed.
- (14) In order to reduce specific fuel consumption in cruise conditions, it was proposed to stop one of the turbo-shaft engines and use the remaining engines for flying. In this way, the one or the only turbo-shaft engines remaining in operation deliver more power than it would be delivered if all the turbo-shaft engines were working and can thus operate with respective lower specific fuel consumption.
- (15) However, such a flight mode with a stopped turbo-shaft engine is only possible under determined operating conditions of the helicopter.
- (16) The sector also feels the need to promptly identify whether the helicopter is satisfying the aforesaid operating conditions.
- (17) Furthermore, the sector feels the need to prevent this flight mode and to facilitate quitting this flight mode with a stopped turbo-shaft engine, if the helicopter is outside these operating conditions.
- (18) The sector also feels the need to promptly indicate to the pilot the correct operations to carry out to bring the helicopter back within these operating conditions, in the event that the aforesaid flight mode is required when the helicopter does not satisfy these operating conditions.
- (19) The sector also feels the need to continuously verify whether the helicopter is at risk of quitting these operating conditions if the aforesaid flight mode has been requested and to promptly warn the pilot of such risk.
- (20) Finally, the need is felt to make said flight mode possible with a stopped turbo-shaft engine, while preserving as much as possible the coupling normally existing between the turbo-shaft engines and the transmission. This is in order to be able to implement this flight mode on already existing helicopters, without having to redesign said coupling.
- (21) As an example, document US-A-2016/0237917 describes a method for optimizing the specific fuel consumption of a twin-engine helicopter. The method involves actuating one of the turbo-shaft engines at a speed indicated as “stabilised flight speed” and the other one at a speed indicated as “super-idle at zero power”. In particular, in the “super-idle at zero power” condition, the turbo-shaft engine has the combustion chamber of the gas generator on while trying to keep a speed at minimum regime in order to reduce the operating temperature and fuel consumption of said gas generator.
- (22) US-A-2013/0219905 describes a method for reducing the specific fuel consumption of a helicopter, in which one of the turbo-shaft engines operates at a speed indicated as “stabilised flight speed” and the other one operates at a speed indicated as “over-idle nil power speed”.
- (23) U.S. Pat. No. 10,487,733 B describes a method for controlling a twin-engine helicopter, in which, in cruise conditions, one of the turbo-shaft engines provides the required power while the other one is kept in a so-called “idling” state by using electrical power generated by converting the mechanical energy provided by a “power spool” of the first turbo-shaft.
- (24) US-A-2020/049025 discloses a method for controlling an aircraft according to the preamble of claim 1 and a method for controlling the aircraft according to the preamble of claim 12.
- (25) EP-A-2886456 discloses a method of managing a power plant for a rotary wing aircraft, comprising two main engines, a secondary engine, and a main power transmission gearbox. The main and secondary engines mechanically drive the main power transmission gearbox, so as to rotate a main rotor of the aircraft. The secondary engine delivers two distinct mechanical power levels so that the main and secondary engines together deliver sufficient mechanical power to enable the aircraft to fly, firstly a first secondary mechanical power MPS1 and secondly a second secondary mechanical power MPS2 suitable for compensating for a loss of main mechanical power from at least one main engine.

#### DISCLOSURE OF INVENTION

- (26) The object of the present invention is to realise an aircraft capable of hovering, which allows

the aforesaid need to be satisfied in a simple and economical way.

(27) According to the invention, the aforesaid object is achieved by a method for controlling an aircraft according to claim 1.

(28) The present invention also relates to an aircraft capable of hovering according to claim 12.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) An embodiment is described below for a better understanding of the present invention, provided by way of non-limiting example with reference to the accompanying drawings, wherein:

(2) FIG. 1 illustrates a perspective view of a twin-engine helicopter realised according to the dictates of the present invention;

(3) FIG. 2 is a flow chart which schematically illustrates a method for controlling an aircraft realised according to the dictates of the present invention;

(4) FIG. 3 is a table illustrating some parameters significant of the method of FIG. 2;

(5) FIG. 4 illustrates the trend over time of the power delivered and the rotation speed of the outlet shaft of the turbo-shaft engines of the helicopter of FIG. 1, starting from a first flight condition in which both engines deliver power and during a transient of entry into a second flight condition in which one of the engines does not deliver engine power;

(6) FIG. 5 illustrates the trend over time of the power delivered and the rotation speed of the outlet shaft of the turbo-shaft engines of the helicopter of FIG. 1, starting from the second flight condition and in a transient of exit from the second operating mode and of return to the first flight condition;

(7) FIG. 6 illustrates the time trend of some flight commands during a recovery manoeuvre performed in the aforesaid second flight condition of the helicopter;

(8) FIG. 7 is a functional diagram of some components of the engine system of the helicopter of FIG. 1;

(9) FIG. 8 schematically illustrates in section some details of the engine system of FIG. 7; and

(10) FIG. 9 is a functional diagram of some components of the helicopter of FIGS. 1 to 8.

### BEST MODE FOR CARRYING OUT THE INVENTION

(11) With reference to FIG. 1, 1 indicates a helicopter essentially comprising a fuselage 2 provided with a front nose 5, a main rotor 3 placed at a top of the fuselage 2 and rotatable around an axis A, and a tail rotor 4 carried by a drift protruding from the fuselage 2 on the opposite side of the nose 5 and rotating around an axis B transverse to axis A.

(12) In greater detail, the rotor 3 comprises a hub 9a of axis A and on which a plurality of blades 9b are constrained in a cantilever manner, which extend radially to axis A.

(13) The blades 9b are articulated on the hub 9a so as to be able to vary their orientation with respect to the relative extension axes, so as to adjust the respective pitch angles with respect to the air current.

(14) It should be noted that in the remainder of this description, expressions such as “above”, “below”, “front”, “rear” and the like are used with reference to forward flight or “hovering” conditions of the helicopter 1 illustrated in FIG. 1 and in which the main rotor 3 is arranged above the fuselage 2 and the nose 5 in front of the tail rotor 4.

(15) The helicopter 1 also comprises: a flight control 14a (only schematically illustrated in FIG. 9) which can be actuated by the pilot and known as a “collective pitch” and which can be actuated to collectively vary the pitch angles of the blades 9b, thus causing an increase or a reduction in the lift generated by the helicopter 1; and a flight control 14b (also only schematically illustrated in FIG. 9) which can be actuated by the pilot and known as “cyclic pitch” and which can be actuated to cyclically vary the pitch angle of the blades 9b according to their relative positions with respect to axis A.

(16) The helicopter **1** also comprises (FIG. 7): an engine system **7** adapted to actuate the main rotor **3** and the tail rotor **4**; and a transmission **8** receiving the motion from the engine system **7** and configured to provide mechanical power in output at suitable rotation speeds to a drive shaft of the main rotor **3** and to a drive shaft of the tail rotor **4**.

(17) The engine system **7** comprises, in turn, a pair of engines **10a**, **10b**.

(18) In the case illustrated, the engines **10a**, **10b** are of the turbo-shaft type, of a known type and described below only to the extent necessary for the present invention.

(19) In greater detail, each engine **10a**, **10b** behaves like a gas turbine system carrying out an open Joule-Brayton thermodynamic cycle.

(20) Each engine **10a**, **10b** comprises a relative outlet shaft **11a**, **11b** rotatable around a relative axis C.

(21) Each engine **10a**, **10b** essentially comprises: an air intake (not illustrated) obtained in a side of the fuselage **2**; a compressor (not illustrated) supplied by the air intake with a fresh air flow rate and adapted to compress said air flow rate; a combustion chamber (not illustrated), in which the air flow rate compressed by the compressor reacts with a fuel flow rate generating a high temperature exhaust gas flow rate; and a pair of turbines (only schematically illustrated), inside which the flow rate of high-temperature exhaust gases leaving the combustion chamber expands, driving the compressor and the relative outlet shaft **11a**, **11b** in rotation around the relative axis C.

(22) Each regime of the engine **10a**, **10b** corresponds to respective values of specific fuel consumption, rotation speed of the relative outlet shaft **11a**, **11b** around the axis C and power made available to the relative outlet shaft **11a**, **11b** itself.

(23) With reference to FIG. 7, the outlet shaft **11a**, **11b** of each engine **10a**, **10b** is connected to a respective inlet shaft **12a**, **12b** of the transmission **8** by means of a relative free wheel **13**.

(24) The free wheel **13** of each engine **10a**, **10b** is provided to allow the rotation of the relative inlet shaft **12a**, **12b** of the transmission **8** in the event of the relative outlet shaft **11a**, **11b** stopping.

(25) In other words, the free wheels **13** allow to preserve the operation of the transmission **8** and, therefore, of the main rotor **3** and of the tail rotor **4** thanks to the power delivered by the other engine **10a**, **10b** in case of failure of one of the engines **10a**, **10b** or in autorotation in case of failure of both engines **10a**, **10b**.

(26) With reference to FIG. 8, each free wheel **13** essentially comprises: a wheel **15** radially external to the axis C and operatively connected to the respective outlet shaft **11a**, **11b**; a wheel **16** radially internal with respect to the axis C and connected to the relative inlet shaft **12a**, **12b**; and a wheel **17** radially intermediate between the wheels **15**, **16**.

(27) The wheel **17** of each free wheel **13** comprises, in turn, a plurality of relative angularly spaced stop teeth **18** (only one of which is illustrated in FIG. 8) and a relative spring **19** which elastically locks the relative teeth **18**.

(28) When power is applied to the outlet shaft **11a**, **11b** of each engine **10a**, **10b**, the wheel **15** of the relative free wheel **13** drives the relative wheel **16** and the relative inlet shaft **12a**, **12b** in rotation.

(29) In the case in which the assembly formed by the wheel **16** and by the relative inlet shaft **12a**, **12b** of the transmission **8** rotates at a rotation speed greater than the rotation speed of the relative outlet shaft **11a**, **11b**—as it occurs, for example, in case of engine sudden stoppage—the relative free wheel **13** allows the disengagement of the relative inlet shaft **12a**, **12b** from the corresponding outlet shaft **11a**, **11b**.

(30) The helicopter **1** also comprises (FIG. 9): a plurality of sensors **21** adapted to provide respective measurements of respective flight parameters and indications relating to the state of correct operation/failure of the systems and equipment of the helicopter **1** not forming part of the engine system **7**; a digital control system of the engine system **7** referred to hereinafter as FADEC **22**; a plurality of sensors **23** arranged inside the engine system **7** and adapted to provide a measurement of the angular speed of the engine system **7** itself; and an avionic system **30**.

- (31) The sensors **21** are configured to measure at least the following parameters: the external temperature OAT; the pressure altitude; the height above ground level; the height above sea level; the indicated speed LAS; the density altitude; the altitude above the minimum flight level and the variometric speed; the attitude of the helicopter **1** and the position of the helicopter **1**.
- (32) The avionic system **30** is programmed to: make at least some of the parameters of the sensors **21** available to the FADEC **22**; and acquire significant parameters of the behaviour of the engines **10a**, **10b** and of the commands imposed by the pilot on the main rotor **3** and tail **4** rotor.
- (33) The FADEC **22** is programmed to process, on the basis of the aforesaid parameters, a control law for the engines **10a**, **10b** in terms of fuel flow rate made available to the respective combustion chambers.
- (34) More precisely, the FADEC **22** processes the control law for the engines **10a**, **10b** independently of the avionic system **30**.
- (35) The helicopter **1** has a first operating configuration, hereinafter referred to as the nominal operating configuration, in which (FIG. 4), the engines **10a**, **10b** make available respective power values **P1**, **P2** substantially equal to each other to the respective outlet shafts **11a**, **11b**.
- (36) The sum of the powers **P1** and **P2** is equivalent to the power **P** necessary for the correct operation of the main **3** and tail **4** rotors.
- (37) The helicopter **1** has a second operating configuration, hereinafter referred to as the ACR operating configuration, i.e. Asymmetric Cruise Regime, in which (FIGS. 5 and 6) the engines **10a**, **10b** make available respective power values **P3**, **P4** different from each other to the respective outlet shafts **11a**, **11b**.
- (38) In particular, the power **P4** is nil while the power **P3** is greater than the power **P1** and equal to the power **P**.
- (39) Preferably, in the ACR mode, the consumption of the engine **10b** is minimal.
- (40) In other words, the engine **10b** is not turned off in the ACR mode, but drives the outlet shaft **11b** in rotation without delivering engine power to the outlet shaft **11b** itself.
- (41) The avionic system **30** is operationally connected with the sensors **21**, **23** and with the flight controls **14a**, **14b**, and is interfaced with the FADEC **22**.
- (42) The avionic system **30** is also configured to continuously monitor the operating conditions of the helicopter **1** and to command, automatically or following a command given by the crew, to the FADEC **22** to place the helicopter **1** in the operating configuration or in the ACR configuration.
- (43) For this purpose, the avionic system **30** essentially comprises: an instrumentation **31** configured to provide the pilot with indications relating to the operating parameters of the helicopter **1** itself; a display device **32** (for example known as CDS) of a series of messages for the crew; a control unit **33**, also known as the “Aircraft Management System—AMS”; and an interface **34** which can be actuated by the crew to command the transition of the helicopter **1** from the nominal configuration to the ACR configuration and vice versa.
- (44) The avionic system **30** further comprises a memory stage **37**.
- (45) In greater detail, the interface **34** comprises (FIG. 2): an ACR REQUEST command that can be actuated by the crew and can be moved between an enabled position, in which the transition from the nominal operating mode to the ACR operating mode is requested, and a disabled position, in which the transition from the ACR operating mode to the nominal operating mode is requested; and an ACR ARM command which can be actuated by the crew.
- (46) In greater detail, the ACR ARM command is a monostable command, of the “momentary pushbutton” type in the case illustrated.
- (47) If the helicopter **1** is in the ACR configuration, the actuation of the ACR ARM command produces the emergency exit manoeuvre of the helicopter **1** from the aforesaid ACR configuration, as will become clear from the remainder of this description.
- (48) If helicopter **1** is in the nominal configuration, the sequential actuation of the ACR ARM and ACR REQUEST commands will produce the request of entry of the helicopter **1** into the ACR

configuration.

(49) More specifically, the ACR ARM command is intended to prevent the ACR REQUEST command from accidentally causing the transition to the ACR configuration.

(50) In the case illustrated, the ACR ARM command is arranged on the flight control **14a** of the collective pitch.

(51) The avionic system **30** is programmed to produce a variation in the state of the FADEC **22**, following the actuation of the ACR ARM and ACR REQUEST commands.

(52) The FADEC **22** also confirms the receipt of the command and the consequent change of state.

(53) The helicopter **1** also comprises an electrical power generation and distribution system **100** (FIG. 7) also known as EPGDS (Electrical Power Generation and Distribution System).

(54) The system **100** essentially comprises, for each engine **10a**, **10b**, a starter **101a**, **101b** adapted to produce the start of the engine **10a**, **10b** itself and an electric current generator **102a**, **102b** actuated by the engine **10a**, **10b** itself.

(55) In the case illustrated, the generators **102a**, **102b** are connected to respective auxiliary transmissions **110a**, **110b** of the respective engines **10a**, **10b**. The auxiliary transmissions **110a**, **110b** are connected to the respective engines **10a**, **10b** in parallel to the respective shafts **12a**, **12b** of the transmission **8**.

(56) The system **100** further comprises, for each engine **10a**, **10b**: a unit **103a**, **103b** known as REPU (Remote Electrical Power Distribution) for the distribution of electrical power to the primary loads essential for the operability of the helicopter **1**; and a unit **104a**, **104b** for the distribution of electrical power to the secondary loads not essential for the operability of the helicopter **1**.

(57) When the helicopter **1** is in the nominal operating configuration, the generators **102a**, **102b** are both active and electrically supply the units **103a**, **104a**; **103b**, **104b**, respectively.

(58) When the helicopter **1** is in the ACR operating configuration, the electrical system **100** is reconfigured so that the generator **102b** connected to the engine **10b** is deactivated and only the electrical loads essential to the operability of the helicopter **1** are supplied only by the generator **102a** actuated by the engine **10a**.

(59) Preferably, the non-essential secondary loads of the unit **104b** are partly supplied by the generator **102a**.

(60) Otherwise, the unit **104b** is disabled.

(61) The display device **32** is programmed to display: an ACR READY signal representative of the fact that the helicopter **1** is in the conditions to operate in the ACR operating configuration; an ACR ARMED signal representative of the fact that the ACR ARM command is in the enabled position; a 1(2) ACR TRANSITION signal representative of the fact that the ACR REQUEST command has been moved to the enabled position and the avionic system **30** is proceeding with the reconfiguration of the electrical system **100**; and an ACR signal representative of the fact that the ACR REQUEST command has been moved to the enabled position, the electrical system **100** has been reconfigured, and the FADEC **22** is proceeding with the reconfiguration of the engines **10a**, **10b**.

(62) The transition from the nominal configuration to the ACR configuration of the helicopter **1** can be activated alternatively on one of the engines **10a**, **10b**.

(63) The control unit **33** of the avionic system **30** is programmed to detect a series of parameters associated with the operating conditions of the helicopter **1**, and to enable the transition of the helicopter **1** from the nominal configuration to the ACR configuration, when said parameters assume respective first values.

(64) The term “operating conditions” in the remainder of this description means both the first operating flight parameters of the helicopter **1** and the second parameters representative of the operating conditions of the systems and equipment of the helicopter **1** essential for the correct operation of the helicopter **1** itself.



- (65) In greater detail (FIGS. 2 and 9), the avionic system **30** is programmed to attribute a TRUE or FALSE value to an “ACR Helicopter Availability” parameter, based on the first parameters detected by the sensors **21**, and on the content of the memory stage **37**.
- (66) In particular, the avionic system **30** is programmed to attribute the TRUE value to an “ACR Helicopter Availability” parameter when the helicopter **1**, with the exception of the engine system **7**, is in a condition to be operated in ACR configuration. Otherwise, the avionic system **30** is programmed to attribute the FALSE value to the “ACR Helicopter Availability” parameter.
- (67) The FADEC **22** is also programmed to evaluate whether the engine system **7** is in a condition to operate in ACR configuration or not.
- (68) In greater detail, the FADEC **22** is configured to attribute TRUE value to an “Engine Availability” parameter if the engine system **7** is in a condition to operate in ACR configuration. Otherwise, the FADEC **22** is programmed to attribute FALSE value to the aforesaid “Engine Availability” parameter.
- (69) The FADEC **22** is also configured to attribute TRUE value to an “ACR Readiness” parameter to the avionic system **30**, when both the “Engine Availability” and “ACR Helicopter Availability” signals assume TRUE value.
- (70) Alternatively, the avionic system **30** is configured to attribute TRUE value to the aforesaid “ACR Readiness” parameter, when both the “Engine Availability” and “ACR Helicopter Availability” signals assume TRUE value.
- (71) Otherwise, the FADEC **22**—or the avionic system **30**—is configured to attribute FALSE value to the aforesaid “ACR Readiness” parameter, when at least one of the “Engine Availability” and “ACR Helicopter Availability” signals assumes FALSE value.
- (72) The avionic system **30** is programmed to display the “ACR READY” signal on the display device **32**, when the FADEC **22**—or the avionic system **30**—attributes TRUE value to the “ACR Readiness” parameter.
- (73) The avionic system **30** is also programmed to allow the pilot to actuate the ACR ARM command through the interface **34** and to place the ACR REQUEST command in the enabled position, when the FADEC **22** attributes TRUE value to the “ACR Readiness” parameter.
- (74) The avionic system **30**, or alternatively the FADEC **22**, are also programmed to make the actuation of the ACR ARM command ineffective, when the FADEC **22** does not make the “ACR Readiness” signal available.
- (75) The FADEC **22** is programmed to request, following the input of ACR ARM and ACR REQUEST commands, the system **100** to reconfigure itself by deactivating the generator **103** and to command the transition of the engines **10a**, **10b** from the nominal configuration to the ACR configuration.
- (76) Preferably, the FADEC **22** is programmed to verify the successful reconfiguration of the system **100** before causing the transition of the engines **10a**, **10b** from the nominal configuration to the ACR configuration. In particular, the aforesaid verification takes place by monitoring the current and electrical voltage values of the generator **103b** provided by the avionic system **30** to the FADEC **22**.
- (77) More precisely, the aforesaid first parameters are representative of the following conditions: i) helicopter **1** is in a determined flight envelope necessary for the helicopter **1** to be able to operate in the ACR configuration; ii) the helicopter **1** is in an area free from obstacles or flight restrictions; iii) the angular speed of the rotor **3** lies in a determined interval of values necessary for the helicopter **1** to be able to operate in the ACR configuration; iv) the indicated speed TAS is in a determined interval necessary for the helicopter **1** to be able to operate in the ACR configuration; and v) the ascent rate and the torque delivered by the engine **10b** are in respective intervals ensuring reduced fuel consumption.
- (78) The avionic system **30** is programmed to compare the data detected by the sensors **21**, **23** with the aforesaid intervals stored in the memory stage **37**.

(79) More precisely, the control unit **33** is programmed to attribute TRUE value to the “ACR Helicopter Availability” signal when the conditions previously identified as i), ii), iii), iv) and v) occur.

(80) In particular, condition i) occurs when: the external temperature OAT is comprised between a first design value and a second design value; the pressure altitude is comprised between a third design value and a fourth design value; the density altitude is comprised between a fifth value and a sixth value.

(81) Condition ii) occurs when: the height above ground level is preferably comprised between 50 and 1000 feet; and the height above sea level is preferably comprised between 50 and 1000 feet.

(82) Preferably, condition ii) is also verified when the altitude above the minimum flight level is preferably comprised between 50 and 1000 feet.

(83) Condition iii) occurs when the angular speed  $N_r$  of the main rotor **3** is comprised between a seventh minimum value and an eighth maximum value, where the seventh value is preferably equal to between 0.85 and 1 times the nominal angular speed of the rotor **3** and the eighth value is preferably comprised between 1 and 1.15 times the aforesaid nominal angular speed and of the rotor **3**.

(84) The term “nominal angular speed” of the rotor **3** in the present description means the angular speed at which the rotor **3** is normally actuated during the operation of the helicopter **1**.

(85) Condition iv) occurs when the indicated speed IAS is comprised between a ninth value and a tenth value, where the ninth value is preferably comprised between 0.5 and 1.5 times the speed in level flight for which the minimum power is required at operating level and temperature and the tenth value is preferably comprised between 1 and 2.5 times the aforesaid speed in level flight for which minimum power is required.

(86) Condition v) occurs when: the torque delivered by the engines **10a**, **10b** is comprised between an eleventh value and a twelfth value, where the eleventh value is preferably comprised between 0.5 and 1 times the torque necessary for a straight level flight at the aforesaid speed for which the minimum power is required and the twelfth value is preferably comprised between 1 and 3 times the aforesaid torque necessary for a straight level flight at the aforesaid speed for which minimum power is required.

(87) Preferably, condition v) is also verified when the variometric speed VS is comprised between a thirteenth value and a fourteenth value, where the thirteenth value is preferably comprised between 0.1 and 1 times the vertical speed of autorotation and the fourteenth value is comprised preferably between 0.1 and 1 times the maximum variometric speed at the aforesaid speed for which minimum power is required.

(88) The aforesaid second parameters are also indicative of the fact that the sensors **21**, **23** do not detect failures incompatible with the ACR configuration and that the operating conditions of the subsystems and plants are not incompatible with the ACR configuration. Examples of such failures incompatible with the ACR configuration are failures of on-board systems necessary to deliver electrical power to aircraft **1** or the functions of the Automatic Flight Control System—hereinafter referred to as AFCS—necessary to help the pilot.

(89) Non-limiting examples of the failures identifiable by the avionic system **30** are as follows: loss of the parameters indicative of whether the helicopter **1** is in the aforesaid flight envelope or not; loss of the parameters identifying the operation of the electrical system **100** and of the engines **10a**, **10b**; loss of the parameters relating to any faults of the engines **10a**, **10b**; errors identifiable by the pilot in the parameters relating to engines **10a**, **10b** and to the electrical system **100**, including false activations of warning signals relating to faults; loss or incorrect supply, identifiable by the pilot, of parameters essential to the flight of the helicopter **1**, for example relating to the hydraulic system or transmission **8**; generation of warning messages that require immediate intervention by the pilot; and failures of the generator **102a** which may produce the shutdown or loss of operating capacity.

(90) It is important to underline that the expression “loss of parameters” used previously is used to

indicate both the unavailability of such parameters on the display device **32** or in the FADEC **22** and the failure to detect such parameters by sensors **21**, **23** and the failure in a unit interposed between the sensors **21**, **23** and the display device **32** or in the FADEC **22**.

(91) The avionic system **30** is also programmed to command the FADEC **22** to transition the helicopter **1** between the nominal and ACR operating configurations.

(92) In greater detail, the avionic system **30** is configured to command the transition of the helicopter **1** from the nominal operating configuration to the ACR operating configuration when both “ACR Helicopter Availability” and “ACR Readiness” parameters assume the TRUE value and when the ACR REQUEST command is in the activated position and the ACR ARM command has been previously actuated.

(93) More precisely, during the transition of the helicopter **1** from the nominal operating configuration to the ACR operating configuration, the avionic system **30** preferably produces a time pause, ten seconds in the case illustrated, before placing the engine **10b** in the condition in which it delivers power **P4**.

(94) The avionic system **30** is programmed to: command a reconfiguration step of the electrical system **100**; and command the FADEC **22** to perform a reconfiguration step of the engines **10a**, **10b**.

(95) This reconfiguration of the engines **10a**, **10b** comprises, in particular: the closure of the valves interposed on the air bleeding line of the compressor of the engine **10b**; the deactivation of the generator **102b** actuated by the engine **10b**; and the reconfiguration of the system **100** so that some of the electrical loads of the units **103b**, **104b** are supplied by the generator **102a** connected to the engine **10a**.

(96) In greater detail, the electrical loads essential for the operability of the helicopter **1** of the units **103b** are electrically connected to the generator **102a** alone actuated by the engine **10a** and the non-essential secondary electrical loads are deactivated or also at least partially electrically connected to the generator **102a** alone, during this reconfiguration step.

(97) Preferably, the aforesaid reconfiguration takes place automatically.

(98) The display device **32** is also commanded by the avionic system **30** to display the ACR signal and to remove the 1(2) ACR TRANSITION signal, at the end of the aforesaid reconfiguration step.

(99) In greater detail, with reference to FIG. **4**, the avionic system **30** is configured, starting from an instant  $t_0$  in which the reconfiguration of the system **100** has been completed and up to an instant  $t_1$  in which the transition from the nominal configuration to the ACR configuration has been completed, in order to request FADEC **22** to: increase the power of the engine **10a** from the value **P1** to the value **P3** reached at the instant  $t_1$ ; decrease the power of the engine **10b** from the value **P2** to the value **P4** reached at the instant  $t_1$ ; and keep the power of the engines **10a**, **10b** at the respective values **P3**, **P4** following the instant  $t_1$  while the helicopter **1** is operating in the ACR configuration.

(100) It is important to underline that the value **P4** is greater than zero. Therefore, the engine **10b** is turned on, when the helicopter **1** is operating in the ACR configuration.

(101) With reference to FIG. **4**, the avionic system **30** is also configured, starting from the instant  $t_0$  in which the angular speeds of the outlet shafts **11a**, **11b** are equal to each other and equal to the value  $n_1$ , to request the FADEC **22** to: decrease the angular speed of the outlet shaft **11b** from the value  $n_1$  to the value  $n_2$  lower than the value  $n_1$ , during the time interval comprised between the instants  $t_0$ ,  $t_1$ ; keep the angular speed of the outlet shaft **11a** at the value  $n_1$  during the time interval comprised between the instants  $t_0$ ,  $t_1$ ; and keep the angular speeds of the outlet shafts **11a**, **11b** at the respective values  $n_1$ ,  $n_2$  following the time instant  $t_1$ .

(102) In particular, following the instant  $t_1$ , the free wheel **13** allows the disengagement between the outlet shaft **11b** rotating at the angular speed  $n_2$  and the relative inlet shaft **12b** to the transmission **8**. In particular, the angular speed of the inlet shaft **12b** is kept constant by the increase in driving torque provided by the inlet shaft **12a** connected to the engine **10a**.

- (103) Preferably, in the event that the ACR REQUEST command is placed in the enabled position when the helicopter **1** is outside the determined flight envelope, the avionic system **30** is configured to assist the crew in a manoeuvre that brings the helicopter **1** within the flight envelope.
- (104) In greater detail, the avionic system **30** is configured to: compare the data detected by the sensors **21**, **23** with the data stored in the memory stage **37** of the avionic system **30** itself; identify the detected data that are outside the aforesaid envelope; and indicate the aforesaid detected data to the crew by means of the display device **32**.
- (105) Preferably, the control unit **33** of the avionic system **30** is programmed to: continuously acquire from the sensors **21**, **23** at least some parameters of the actual flight envelope of the helicopter **1** and monitor that at least these parameters of the actual flight envelope are within the determined envelope of determined flight, when the helicopter **1** is in the ACR configuration; obtain a trend of said parameters; and display on the display device **32** a warning message representative of the fact that the trend of said parameters produces the exit of the helicopter **1** from the aforesaid determined flight envelope.
- (106) In particular, the aforesaid trend is obtained constantly and for any operating configuration of the helicopter **1** itself.
- (107) In greater detail, the parameters of the actual flight envelope acquired by the avionic system **30** comprise at least the indicated speed LAS, the vertical speed VS and the altitude ALT.
- (108) More precisely, the avionic system **30** is programmed to detect a trend of the vertical speed and compare it with the ascent rate of the determined flight envelope.
- (109) The avionic system **30** is also programmed to detect the altitude trend, compare it with a trend obtainable from a database stored in the stage **37** and with the level of the determined flight envelope.
- (110) In particular, the aforesaid database comprises, for each latitude and longitude value, an altitude value above the sea.
- (111) The avionic system **30** is also configured to produce the transition of the helicopter **1** from the ACR operating configuration to the nominal operating configuration in the following circumstances: to perform a commanded nominal emergency exit from the ACR operating configuration; to perform a commanded emergency exit from the ACR configuration; to perform an automatic emergency exit from the ACR configuration, when the “ACR Helicopter Availability” parameter assumes the FALSE value; and to perform an automatic nominal exit, in case of minor anomalies of the helicopter **1** which do not require an automatic emergency exit.
- (112) Preferably, the commanded nominal commanded exit manoeuvre and the automatic nominal exit manoeuvre take place in longer times than the commanded emergency exit manoeuvres and the automatic emergency exit.
- (113) In greater detail, the control unit **33** of the avionic system **30** is programmed to request the FADEC **22** to perform a commanded nominal exit from the ACR configuration, when the crew moves the ACR REQUEST command to the deactivated position, in the ACR configuration of the helicopter.
- (114) Following the movement of the ACR REQUEST command to the deactivated position, the display device **32** is programmed to: display the 1(2) ACR TRANSITION indication, following approval by the FADEC **22**; remove the ACR indication, once both engines **10a**, **10b** deliver the same powers P1, P2; and remove the 1(2) ACR TRANSITION indication and display the ACR ARMED indication, once helicopter **1** is outside the ACR configuration.
- (115) Alternatively, following the movement of the ACR REQUEST command to the deactivated position, the display device **32** is programmed to remove the indication, once the engines **10a**, **10b** have the same outlet temperatures as the hot gases from the relative turbines.
- (116) The control unit **33** of the avionic system **30** is also programmed to perform the commanded emergency exit manoeuvre when the ACR ARM command is actuated with the helicopter **1** in the ACR operating configuration.

(117) Under such conditions, the display device **32** is programmed to: display the 1(2) ACR TRANSITION indication following the confirmation by the FADEC **22** and during the re-configuration of the engines **10a**, **10b**; remove the ACR indication, once both engines **10a**, **10b** deliver the same powers **P1**, **P2**; and remove the 1(2) ACR TRANSITION indication and display the ACR READY indication, once helicopter **1** is outside the ACR configuration.

(118) The display device **32** is also programmed during the automatic emergency exit from the ACR configuration to: display the 1(2) ACR TRANSITION indication; remove the ACR indication, once the engine **10b** delivers the same power as the engine **10a**; and remove the 1(2) ACR TRANSITION indication once the reconfiguration of the helicopter **1** has been completed.

(119) The control unit **33** of the avionic system **30** is also programmed to perform the automatic nominal exit manoeuvre in case of minor anomalies of the helicopter **1** which do not require an automatic emergency exit.

(120) During this automatic nominal exit manoeuvre, the display device **32** is programmed to: display the 1(2) ACR TRANSITION indication, following approval by the FADEC **22**; remove the ACR indication, once both engines **10a**, **10b** deliver the same powers **P1**, **P2**; remove the 1(2) ACR TRANSITION indication and display the ACR ARMED indication, once helicopter **1** is outside the ACR configuration.

(121) The avionic system **30** is also programmed to inhibit the ARM REQUEST function and attribute the FALSE value to the ACR ARM parameter, at the end of the completion of the automatic emergency exit manoeuvre.

(122) In the conditions of automatic emergency exit, the avionic system **30** switches the ACR REQUEST command into the respective disabled position and inhibits the enabling thereof.

(123) In the automatic emergency exit conditions, the avionic system **30** monitors the systems of the helicopter **1** and identifies any failures.

(124) With reference to FIG. 5, the FADEC **22** is programmed, starting from an instant  $t_3$  in which the transition from the ACR configuration to the nominal configuration is commanded, to: increase the angular speed of the outlet shaft **11b** from the value  $n_2$  to the value  $n_1$ , during the time interval comprised between the instant  $t_3$  and an instant  $t_4$ ; keep the angular speed of the outlet shaft **11a** at the value  $n_1$  during the time interval comprised between the instants  $t_3$ ,  $t_4$ ; and keep the angular speeds of the outlet shafts **11a**, **11b** at the respective values  $n_1$  equal to each other following the time instant  $t_4$ .

(125) In particular, in the time interval comprised between the time instants  $t_3$ ,  $t_4$ , the engine **10b** accelerates and brings the outlet shaft **11b** to the same angular speed as the relative inlet shaft **12b** to the transmission **8**.

(126) The FADEC **22** is also programmed to: decrease the power of the engine **10a** from the value **P3** to the value **P1** in the time interval comprised between the time instant  $t_4$  and the time instant  $t_5$  in which the aforesaid transition has been completed; and increase the power of the engine **10b** from the value **P4** to the value **P2** reached in the time interval comprised between the time instants  $t_4$ ,  $t_5$ ; and keep the power of the engines **10a**, **10b** at the respective values **P1**, **P2** equal to each other following the instant  $t_5$  while the helicopter **1** is operating in the nominal configuration.

(127) Preferably, the FADEC **22** is programmed during the transition of the helicopter **1** from the ACR operating configuration to the nominal operating configuration under the conditions of commanded nominal exit, automatic nominal exit, automatic emergency exit and commanded emergency exit in order to: produce respective time transients different from each other for reconfiguring the engines **10a**, **10b**; produce different time trends of the sliding speed between the wheels **16**, **15** of the free wheels **13**; and produce different power values made available by the engines **10a**, **10b**, once the respective free wheels **13** allow the angular rotation of the respective outlet shafts **11a**, **11b** and inlet shafts **12a**, **12b**.

(128) The helicopter **1** also comprises an automatic AFCS flight control system programmed to: receive from the FADEC **22** information relating to the fact that the helicopter **1** is transitioning

between the nominal configuration and the ACR configuration; evaluate any corrections for the flight controls **14a**, **14b**; and display the aforesaid corrections to the crew through the display device **32** or directly implement these corrections on the flight controls **14a**, **14b** in order to minimize perturbations in the trajectory and/or attitude of the helicopter **1** and in the angular speed of the rotor **3**.

(129) Preferably, the pilot or the AFCS flight control system acts on the flight controls **14a**, **14b** to perform a recovery manoeuvre in the event of a loss of power of the engine **10a** when the helicopter **1** is in the ACR configuration.

(130) This recovery manoeuvre is adapted to compensate for the loss of transient power in the time interval between when the engine **10a** starts failing and when the engine **10b** is able to deliver the power necessary to sustain the helicopter **1**.

(131) It is thus possible to keep the helicopter **1** within the relative operational and design limits during the aforesaid time interval.

(132) Preferably, the recovery manoeuvre is such as to produce the reduction in the level and speed of the helicopter **1** (FIG. 6). In particular, the recovery manoeuvre produces the arrangement of the helicopter **1** in a diving attitude and the acquisition of a vertical descent speed by reducing the pitch angles of the blades **9b** by means of the flight control **14a**. For example, the aforesaid recovery manoeuvre is an autorotation manoeuvre in an emergency condition.

(133) When the helicopter **1** is in the ACR configuration, the free wheel **13** interposed between the outlet shaft **12b** and the inlet shaft **11b** bears the mechanical loads due to the relative sliding between the respective wheels **15**, **16**. More specifically, the difference in rotation speed between the wheels **16**, **15** is chosen so as to reduce the local pressure loads on the components of the free wheel **13**. Furthermore, the free wheel **13** is normally sized to permit these speed differences with a reduced impact on the fatigue life.

(134) The operation of the helicopter **1** is described starting from a condition in which helicopter **1** is in the nominal configuration and the ACR REQUEST command is in the disabled position (FIG. 2).

(135) Under these conditions, the engines **10a**, **10b** deliver the respective powers **P1**, **P2** and the outlet shafts **11a**, **11b** rotate at respective angular speeds  $n1$  equal to each other integrally with the respective inlet shafts **12a**, **12b** of the transmission **8**.

(136) Both generators **102a**, **102b** are driven by respective engines **10a**, **10b** and electrically supply respective units **103a**, **103b**; **104a**, **104b**.

(137) The control unit **33** of the avionic system **30** acquires from the sensors **21** the first parameters associated with the operating conditions of the helicopter **1** and the second parameters associated with any failures of the systems and equipment of the helicopter **1**.

(138) The control unit **33** compares these first parameters with the values stored in the stage **37**.

(139) In greater detail (FIG. 9), the control unit **33** of the avionic system **30** attributes TRUE or FALSE value to the “ACR Helicopter Availability” parameter, based on the first parameters detected by the sensors **21**, on the content of the memory stage **37**, and on any failures found by the avionic system **30**.

(140) In particular, the control unit **33** of the avionic system **30** attributes TRUE value to the “ACR Helicopter Availability” parameter when the helicopter **1**, with the exception of the engine system **7**, is in a condition to be operated in ACR configuration.

(141) Even more precisely, the control unit **33** of the avionic system **30** attributes TRUE value to the “ACR Helicopter Availability” parameter, when all conditions i), ii), iii), iv) and v) occur and in the absence of significant failures in the helicopter systems **1**.

(142) On the contrary, the control unit **33** attributes FALSE value to the parameter “ACR Helicopter Availability” when the helicopter **1** cannot operate in the ACR configuration.

(143) The FADEC **22** attributes TRUE value to the “Engine Availability” parameter if the engine system **7** is in a condition to operate in ACR configuration.

(144) The FADEC **22**, or alternatively the avionic system **30**, attributes TRUE value to the “ACR Readiness” parameter to the avionic system **30**, when both the “Engine Availability” and “ACR Helicopter Availability” signals assume TRUE value.

(145) Consequently, the display device **32** displays the “ACR READY” parameter.

(146) Under such conditions, the crew can request the transition to the ACR configuration by first actuating the ACR ARM command and then placing the ACR REQUEST command in the enabled position. Following the input of the ACR ARM and ACR REQUEST commands, the avionic system **30** produces the reconfiguration of the system **100** and requests the FADEC **22** to produce the transition of the engines **10a**, **10b** from the nominal configuration to the ACR configuration.

(147) In particular, the system **100** is reconfigured, preferably automatically, so as to deactivate the generator **102b** connected to the engine **10b** and keep the generator **102a** connected to the engine **10a** active in order to supply the units **103a**, **104a**.

(148) The essential electrical loads of the unit **103b** are electrically connected to the generator **102a** alone actuated by the engine **10a** alone and the non-essential electrical loads are disabled during said reconfiguration step.

(149) Preferably, the FADEC **22** verifies that the reconfiguration of the system **100** has occurred before causing the transition of the engines **10a**, **10b** from the nominal configuration to the ACR configuration.

(150) In particular, during the reconfiguration step of the system **100**, the display device **32** displays the 1(2) ACR TRANSITION indication.

(151) During the reconfiguration step of the engines **10a**, **10b**, the display device **32** displays the ACR indication.

(152) Preferably, the reconfiguration of the helicopter **1** begins after a time interval has elapsed since placing the ACR REQUEST command in the enabled position.

(153) At the end of this time interval, the FADEC **22** reduces the fuel supply to the combustion chamber to the engine **10b** and starts the reconfiguration of the engines **10a**, **10b**.

(154) In greater detail, the FADEC **22**, starting from the instant  $t_0$  (FIG. 4), in which the reconfiguration of the system **100** has been completed and up to an instant  $t_1$ , in which the transition from the nominal configuration to the ACR configuration has been completed: increases the power of the engine **10a** from the value  $P_1$  to the value  $P_3$  reached at the instant  $t_1$ ; decreases the power of the engine **10a** from the value  $P_2$  to the value  $P_4$  reached at the instant  $t_1$ ; and keeps the power of the engines **10a**, **10b** at the respective values  $P_3$ ,  $P_4$  following the instant  $t_1$  while the helicopter **1** is operating in ACR configuration.

(155) The FADEC **22**, starting from the instant  $t_0$  in which the angular speeds of the outlet shafts **11a**, **11b** are equal to each other and equal to the value  $n_1$ : decreases the angular speed of the outlet shaft **11b** from the value  $n_1$  to the value  $n_2$  lower than the value  $n_1$ , during the time interval comprised between the instants  $t_0$ ,  $t_1$ ; keeps the angular speed of the outlet shaft **11a** at the value  $n_1$  during the time interval comprised between the instants  $t_0$ ,  $t_1$ ; and keeps the angular speeds of the outlet shafts **11a**, **11b** at the respective values  $n_1$ ,  $n_2$  following the time instant  $t_1$ .

(156) In particular, the inlet shaft **12a** of the transmission **8** is driven in rotation by the outlet shaft **11a** of the engine **10a** delivering the power  $P_1$ . The angular speed of the inlet shaft **12b** is kept constant by the increase in driving torque provided by the inlet shaft **12a** connected to the engine **10a**.

(157) Following the instant  $t_1$ , the free wheel **13** allows the relative rotation with sliding between the inlet shaft **12b** of the transmission **8** and the outlet shaft **11b** of the engine **10b** rotating at the angular speed  $n_1$ .

(158) The avionic system **30** continuously acquires the signals of the sensors **21**, **23** and compares that the operating parameters continue to remain in the intervals stored in the stage **37**.

(159) If some of the operating parameters are outside the intervals stored in the stage **37** and/or failures incompatible with the ACR configuration are found, the system **100** and the engines **10a**,

**10b** return to the nominal configuration.

(160) When the helicopter **1** operates in the ACR configuration, the control unit **33** monitors that some parameters of the actual flight envelope of the helicopter **1** remain within the determined envelope of determined flight, and derives a trend of said parameters.

(161) The display device **32** provides the pilot with a warning message relating to the fact that the trend of said parameters produces the exit of the helicopter **1** from the aforesaid determined flight envelope.

(162) In case of loss of power of the engine **10a** when the helicopter **1** is in the ACR configuration, the pilot or the AFCS flight control system act on the flight controls **14a**, **14b** to perform a recovery manoeuvre, in order to compensate for the loss of transient power in the time interval between when the engine **10a** starts failing and when the engine **10b** is able to deliver the power necessary to sustain the helicopter **1**.

(163) Preferably, the recovery manoeuvre produces the reduction in the level and speed of the helicopter **1** (FIG. 6). In particular, the recovery manoeuvre produces the arrangement of the helicopter **1** in a diving attitude and the acquisition of a vertical descent speed by reducing the pitch angles of the blades **9b** by means of the flight control **14a**.

(164) The transition of the helicopter **1** from the ACR configuration to the nominal configuration takes place via a commanded nominal exit or a commanded emergency exit or an automatic emergency exit or an automatic nominal exit.

(165) In particular, the crew produces the commanded nominal exit by placing the ACR REQUEST command in the disabled position.

(166) In response to this deactivation of the ACR REQUEST command, the display device **32** displays the 1(2) ACR TRANSITION indication, following approval by the FADEC **22**; removes the ACR signal, once both engines **10a**, **10b** deliver the same powers **P1**, **P2**; removes the 1(2) ACR TRANSITION indication; and displays the ACR ARMED indication once the helicopter **1** is in the nominal configuration.

(167) The crew produces the commanded emergency exit by actuating the ACR ARM command.

(168) In response to said actuation, the display device **32** displays the 1(2) ACR TRANSITION indication following the confirmation by the FADEC **22**, removes the ACR indication, once both engines **10a**, **10b** deliver the same powers **P1**, **P2**; removes the 1(2) ACR TRANSITION indication once the helicopter **1** is outside the ACR configuration; and displays the ACR READY indication.

(169) The avionic system **30** places the ACR REQUEST command in the deactivated position, thus preventing the possibility of returning to the ACR configuration.

(170) The control unit **33**, or alternatively the FADEC **22** itself, commands the FADEC **22** to produce the automatic emergency exit from the ACR configuration, when one of the “ACR Helicopter Availability” and “Engine Availability” parameters assumes FALSE value, that is, when at least one of the conditions i), ii), iii), iv), v) is not verified.

(171) Under such circumstances, the display device **32** displays the 1(2) ACR TRANSITION indication, removes the ACR indication once the engine **10b** delivers the same power as the engine **10a**, and removes the 1(2) ACR TRANSITION indication once the reconfiguration of the helicopter **1** has been completed.

(172) The control unit **33** resets the ACR armed state, does not display the ACR ARMED indication and places the ARM REQUEST command in the disabled position, at the end of the completion of the automatic emergency exit manoeuvre.

(173) The display device **32** displays the 1(2) ACR TRANSITION indication during the transition from the ACR configuration to the nominal configuration of the helicopter **1** and removes the ACR indication at the end of the aforesaid transition.

(174) Under the automatic emergency exit conditions, the sensors **21** recognize the presence of one or more failures incompatible with the ACR configuration, monitor the systems of the helicopter **1** with the exception of the engines **10a**, **10b**, and communicate this failure to the avionic system **30**.



- (175) The FADEC **22** is commanded to implement a control law of the outlet shaft **11b** of the engine **10b** so as to ensure correct power values to the inlet shaft **12b** of the transmission **8**.
- (176) The avionic system **30** is also programmed to produce a rapid activation of the aids to the pilot to reconfigure the system **100**.
- (177) The control unit **33** requests the FADEC **22** to perform the automatic nominal exit manoeuvre, in case of minor anomalies of the helicopter **1** that do not require an automatic emergency exit.
- (178) In greater detail, the control unit **33** also requests the FADEC **22**, starting from the instant  $t_3$  in which the transition from the ACR configuration to the nominal configuration is commanded, to: increase the angular speed in the outlet shaft **11b** from the value  $n_2$  to the value  $n_1$ , during the time interval comprised between the instant  $t_3$  and an instant  $t_4$ ; keep the angular speed of the outlet shaft **11a** at the value  $n_1$  during the time interval comprised between the instants  $t_3$ ,  $t_4$ ; and keep the angular speeds of the outlet shafts **11a**, **11b** at the respective values  $n_1$  equal to each other following the time instant  $t_4$ .
- (179) In particular, in the time interval comprised between the time instants  $t_3$ ,  $t_4$ , the engine **10b** accelerates and brings the outlet shaft **11b** to the same angular speed as the relative inlet shaft **12b** to the transmission **8**.
- (180) The control unit **33** also commands the FADEC **22** to: decrease the power of the engine **10a** from the value  $P_3$  to the value  $P_1$  in the time interval comprised between the time instant  $t_4$  and the time instant  $t_5$  in which the aforesaid transition has been completed; increase the power of the engine **10a** from the value  $P_4$  to the value  $P_2$  reached in the time interval comprised between the time instants  $t_4$ ,  $t_5$ ; and keep the power of the engines **10a**, **10b** at the respective values  $P_1$ ,  $P_2$  equal to each other following the instant  $t_5$  while the helicopter **1** is operating in the nominal configuration.
- (181) Depending on the type of manoeuvre for exiting the ACR configuration, the control unit **33** commands the FADEC **22** to produce different time trends of the sliding speed between the wheels **16**, **15** of the free wheels **13**, different time trends of the sliding speed between the wheels **16**, **15** of the free wheels **13**, and to produce different power values made available by the engines **10a**, **10b** once the respective free wheels **13** allow the angular rotation of the respective outlet shafts **11a**, **11b** and inlet shafts **12a**, **12b**.
- (182) For each of the aforesaid exit manoeuvres, the generator **102b** connected to the engine **10b** is activated again as well as the units **103b**, **104b** electrically connected to the generator **102b** itself, at the end of the transition from the ACR configuration to the nominal configuration of the helicopter **1**.
- (183) From an examination of the characteristics of the helicopter **1** according to the present invention, the advantages that it allows obtaining are evident.
- (184) In particular, the control unit **33** of the avionic system **30** is programmed to: acquire a series of parameters associated with the operating conditions of the helicopter **1**; and enable the transition from the nominal configuration to the ACR configuration, when the aforesaid parameters assume respective first values.
- (185) In this way, it is possible to ensure that the helicopter **1** operates in the ACR configuration with lower specific fuel consumption, only when it is in a determined flight envelope and in the absence of failures of essential components of the helicopter **1** itself that affect the operativeness thereof in the ACR configuration.
- (186) This ensures that the engine system **7** can deliver maximum power when required by the operating conditions of the helicopter **1** and can minimize its consumption when permitted by the operating conditions of the helicopter **1** itself.
- (187) Thanks to the fact that the control unit **33** continuously acquires the parameters associated with the operating conditions of the helicopter **1**, it is possible to perform an automatic emergency exit from the ACR configuration to the operating configuration, as soon as the helicopter **1** is

outside the determined envelope of determined flight or there are failures of the helicopter **1**.

(188) When the helicopter **1** operates in the ACR configuration, the control unit **33** monitors that some parameters of the actual flight envelope of the helicopter **1** are within the determined envelope of determined flight, and derives a trend of these parameters.

(189) The display device **32** provides the pilot with a warning message relating to the fact that the trend of said parameters produces the exit of the helicopter **1** from the aforesaid determined flight envelope.

(190) It is thus possible to promptly inform the crew that helicopter **1** is approaching a flight envelope in which the ACR configuration is not allowed.

(191) In case of loss of power of the engine **10a** when the helicopter **1** is in the ACR configuration, the pilot or the AFCS flight control system act on the flight controls **14a**, **14b** to perform a recovery manoeuvre adapted to compensate for the transient power loss in the time interval between when the engine **10a** starts failing and when the engine **10b** is able to deliver the power necessary to sustain the helicopter **1**.

(192) It is thus possible to keep the helicopter **1** within the relative operational and design limits during the aforesaid time interval.

(193) The control unit **33** of the avionic system **30** also allows: a commanded nominal exit from the ACR operating configuration, should the crew consider it necessary; and/or an automatic non-emergency exit from the ACR operating configuration in case of minor anomalies of the helicopter **1** that do not require an automatic emergency exit.

(194) In greater detail, the commanded nominal exit manoeuvre can be requested by the crew even when the parameters assume the first values.

(195) In this way, it is possible to exit the ACR configuration due to a need encountered by the crew, for example because the performance in this mode is not considered adequate for the specific flight conditions and the mission, or because anomalies not included among those automatically detected by the avionic system **30** requires the manual configuration to be restored.

(196) It is thus possible to guarantee the maximum flexibility of use of the helicopter **1** along with the possibility of reducing consumption in the ACR configuration.

(197) It is important to underline that in the ACR configuration, the engine **10b** is not turned off, but delivers substantially nil power **P2**.

(198) Consequently, thanks to the fact that the engine **10b** remains active also in the ACR configuration, the time necessary to place the helicopter **1** in the nominal configuration in which the engine **10b** delivers the power **P2** is particularly reduced compared to known solutions in which one of the engines has been completely turned off.

(199) In the ACR configuration of the helicopter **1**, the outlet shaft **11b** of the engine **10b** rotates at an angular speed  $n_1$  lower than the angular speed  $n_2$  of the inlet shaft **12b** of the transmission, thanks to the presence of the free wheel **13**.

(200) In other words, the free wheel **13** allows to support the mechanical loads due to the relative sliding between the wheel **16** and the wheel **15** which last over the entire time in which the helicopter **1** is in the ACR configuration.

(201) In this way, the same free wheel **13** normally intended to allow the operation of the rotor **3** to be preserved for a short time in the event of exceptional failure of the engine **10b** is also advantageously used to allow the operation of the rotor **3** in the ACR configuration over a prolonged period of time.

(202) This is possible since the difference in rotation speed between the wheels **16**, **15** is chosen in such a way as to reduce the local pressure loads on the components of the free wheel **13**, and since the free wheel **13** is normally sized to allow such differences in speed with a reduced impact on fatigue life.

(203) Finally, it is clear that modifications and variations can be made to the aircraft **1** previously described without thereby departing from the scope of protection of the present invention.

(204) In particular, the aircraft capable of hovering could be a convertiplane instead of the helicopter **1**.

(205) More specifically, the aircraft capable of hovering could be a multirotor aircraft, for example with two coaxial rotors or of the intermeshing type.

(206) Condition ii) could be verified not by the avionic system **30**, but by the pilot under manual flight conditions.

(207) The engines **10a**, **10b** could be heat engines different from the turbo-shaft engines or electric engines actuated by batteries or hybrid engines.

## Claims

1. A method for controlling a hovering aircraft which includes a first engine, a second engine, a rotor operatively connected to the first and second engines, and a transmission interposed between the first and second engine and the rotor, the transmission having a first inlet connected to the first engine and a second outlet coupled to the second engine the method comprising: a) measuring a series of parameters associated with the operating conditions of said aircraft, the parameters including: i) an external temperature (OAT); ii) an external pressure, an external air density; iii and iv) heights of the aircraft above ground level and sea level; v) an angular speed of the rotor; vii) an indicated speed; viii) a torque delivered by the first and second engines; ix) an altitude above a minimum flight level; and x) a variometric speed (Vz), b) determining whether the measured parameters indicate one or more of the following conditions: i) the external temperature (OAT) is between a first operating value and a second operating value; ii) the external pressure is between a third operating value and a fourth operating value; iii) the external air density is between a fifth operating value and a sixth operating value; iv) a height above ground level is between 50 and 1000 feet; v) the height above sea level is between 50 and 1000 feet; vi) the angular speed of the rotor is between 0.85 and 1.15 times a nominal angular speed; vii) the indicated speed (IAS) is between 0.5 and 2.5 times a first speed in level flight for which the minimum power at the torque delivered by the first and second engines level and operating temperature is required; viii) is between 0.5 and 3 times a first torque necessary for a straight level flight at the first speed; ix) an altitude above a minimum flight level is between 50 and 1000 feet; and a variometric speed (Vz) is between 0.1 and 1 times a vertical speed of autorotation and 0.1 and 1 times a maximum variometric speed at the first speed, c) when it is determined that the conditions i) through ix) are fulfilled: c1) transitioning operation of the aircraft from a first configuration in which both the first engine is operating with first power (P1) and the second engine is operating with second power (P2) to a second configuration in which the first engine is made to operate a third power (P3) greater than the first power (P1) and the second engine is made to operate at a nil power value (P4), wherein the third power being equal to the sum of first power (P1) and second power (P2); d) when it is determined that conditions i) through ix) are not fulfilled and a difference between the measured parameters and at least one condition i) through ix) is greater than a minor anomaly, and the aircraft is operating in the second configuration: d1) performing a first automatic transition operating of the aircraft from the second configuration to the first configuration, e) when it is determined that conditions i) through ix) are not fulfilled and a difference between the measured parameters and at least one condition i) through ix) is within a tolerance considered to minor anomaly, and the aircraft is operating in the second configuration: e1) performing a second automatic transition of the aircraft from said second configuration to the first configuration, wherein the second automatic transition is slower than the first automatic transition.

2. The method according to claim 1, characterized in that at least some of said parameters define a flight envelope of said aircraft (**1**); and/or at least further said parameters are indicative of a failure of said aircraft (**1**).

3. The method according to claim 1, characterized in that it comprises step iv) of preventing the

transition of said aircraft (1) from said first configuration to said second configuration, when at least one of said parameters have a second value different from said first value; and/or characterized in that it comprises step vi) of performing a commanded emergency transition of said aircraft (1) from said second configuration to said first configuration, when said parameters have respective first values and following a first command.

4. The method according to claim 1, characterized in that it comprises the steps of: vii) producing a trend of at least some of said parameters, when said aircraft (1) is in said second configuration; and viii) generating a warning signal if said trend leads said at least some parameters to have second values different from said first values.

5. The method according to claim 1, characterized in that it comprises the steps: ix) detecting that said first engine (10a) starts failing; x) reducing the level and the vertical speed of said aircraft (1); and xi) increasing the power delivered by said second engine (10b).

6. The method according to claim 1, characterized in that it comprises step xv) of controlling said second engine (10b) so that the relative second outlet member (12) rotates at an angular speed greater than zero in said second configuration.

7. The method according to claim 1, characterized in that it comprises step xvi) producing a relative angular sliding between said second outlet member (11b) and said second inlet (12b) by means of a one-way clutch (13) when said aircraft (1) is in said second configuration and in case of failure of said second engine (10b); said one-way clutch (13) comprising a first element (16) rotatable integrally with said relative second inlet (12b) and a second element (15) rotatable integrally with said second outlet shaft (11b); said clutch (13) rotationally decoupling said second inlet (12b) and second outlet shaft (11b), when said first element (16) rotates at a lower speed than said second element (15), and rotationally coupling said second inlet (12b) and second outlet shaft (11b) when said first and second element (16, 15) rotate at the same angular speed.

8. The method according to claim 7, characterized in that it comprises the steps of: xvii) producing different time trends of the sliding speed between said first element (16) and second element (15), according to the type of transition from said second configuration to said first configuration; and/or xviii) producing different power values (P1, P2) made available respectively by said first engine (10a) and second engine (10) once said one-way clutch (13) couples said second inlet (12b) and second outlet shaft (11b), according to the type of transition from said second configuration to said first configuration.

9. The method according to claim 1, characterized in that it comprises the steps of: xix) mechanically connecting a first electrical generator (102a) to said first outlet member (11a) of said first engine (10a); xx) electrically supplying first electrical loads (103a, 104a) with said first electrical generator (102a), when said aircraft (1) is in said first configuration; xxi) mechanically connecting a second electrical generator (102b) to said second outlet member (11b) of said second engine (10b); xxii) electrically supplying second electrical loads (103b, 104b) with said second electrical generator (102b), when said aircraft (1) is in said first configuration; xxiii) deactivating said second generator (102b) and the non-essential portion of said second electrical loads (103b, 104b), and supplying at least an essential portion of said second electrical loads with said first generator (102a), when said aircraft (1) is in said second configuration.

10. A hovering aircraft comprising: a first engine; a second engine; at least one rotor operatively connected to the first engine and to the second engine; a transmission interposed between the first and second engines and the rotor including a first inlet connected to the first engine and a second inlet connected to the second engine; a plurality of sensors to measuring parameters associated with external conditions and conditions of components of the aircraft; a control unit (30, 33, 22) configured to: a) receive a series of sensed parameters from the plurality of sensors, the parameters including: i) an external temperature (OAT); ii) an external pressure, an external air density; iii and iv) heights of the aircraft above ground level and sea level; v) an angular speed of the rotor; vii) an indicated speed; viii) a torque delivered by the first and second engines; ix) an altitude above a

minimum flight level; and x) a variometric speed ( $V_z$ ), b) determine whether the measured parameters indicate one or more of the following conditions: i) the external temperature (OAT) is between a first operating value and a second operating value; ii) the external pressure is between a third operating value and a fourth operating value; iii) the external air density is between a fifth operating value and a sixth operating value; iv) a height above ground level is between 50 and 1000 feet; v) the height above sea level is between 50 and 1000 feet; vi) the angular speed of the rotor is between 0.85 and 1.15 times a nominal angular speed; vii) the indicated speed (IAS) is between 0.5 and 2.5 times a first speed in level flight for which the minimum power at the torque delivered by the first and second engines level and operating temperature is required; viii) is between 0.5 and 3 times a first torque necessary for a straight level flight at the first speed; ix) an altitude above a minimum flight level is between 50 and 1000 feet; and a variometric speed ( $V_z$ ) is between 0.1 and 1 times a vertical speed of autorotation and 0.1 and 1 times a maximum variometric speed at the first speed, c) when it is determined that the conditions i) through ix) are fulfilled: c1) transition operation of the aircraft from a first configuration in which both the first engine is operating with first power ( $P_1$ ) and the second engine is operating with second power ( $P_2$ ) to a second configuration in which the first engine is made to operate a third power ( $P_3$ ) greater than the first power ( $P_1$ ) and the second engine is made to operate at a nil power value ( $P_4$ ), wherein the third power being equal to the sum of first power ( $P_1$ ) and second power ( $P_2$ ); d) when it is determined that conditions i) through ix) are not fulfilled and a difference between the measured parameters and at least one condition i) through ix) is greater than a minor anomaly, and the aircraft is operating in the second configuration: d1) perform a first automatic transition operating of the aircraft from the second configuration to the first configuration, e) when it is determined that conditions i) through ix) are not fulfilled and a difference between the measured parameters and at least one condition i) through ix) is within a tolerance considered to minor anomaly, and the aircraft is operating in the second configuration: e1) perform a second automatic transition of the aircraft from said second configuration to the first configuration, wherein the second automatic transition is slower than the first automatic transition.

11. The aircraft according to claim 10, characterized in that it comprises: a plurality of sensors (**21**, **23**) adapted to detect first said parameters defining a flight envelope of said aircraft (**1**) and second said parameters indicative of a failure of said aircraft (**1**), and functionally connected with said control unit (**30**, **33**); and a FADEC (**22**) configured to command said first and second engine (**10a**, **10b**) and commanded by said control unit (**30**, **33**).

12. The aircraft according to claim 10, characterized in that said control unit (**30**, **33**) comprises a storage stage (**37**) of the said first values; and/or characterized in that it comprises a display device (**32**) commanded by said control unit (**30**, **32**, **33**) and configured to display to the crew: a first signal (ACR READY) representative of the fact that said aircraft (**1**) including said first and second engine (**10a**, **10b**) is in a condition to be operated in ACR configuration; and/or a second signal (ACR ARMED) representative of the fact that said transition from said first to the second configuration is enabled; and/or a third signal (ACR) representative of the fact that said aircraft (**1**) operates, in use, in said second configuration.

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